

# gat-typst Package Manual

Version 0.1.0

*A comprehensive Typst package for academic writing, homework assignments, and cheatsheets*

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# 1 Introduction

The `gat-typst` package is a comprehensive template system for academic writing in Typst, with a focus on mathematics-heavy homework assignments, cheatsheets, and technical documents. It provides opinionated defaults, extensive math utilities, and a curated collection of the best packages from the Typst ecosystem.

This is a growing library for Gatlen Culp's personal usage and is unstable.

## 1.1 Key Features

- **Templates:** Ready-to-use templates for homework, cheatsheets, and general documents
- **Math Utilities:** Enhanced math notation with semantic symbols and custom functions
- **Integrated Packages:** 20+ carefully selected packages for diagrams, code, and formatting
- **Opinionated Styling:** Professional defaults for code, tables, and equations
- **Convenience Functions:** Colored boxes, custom math operators, and more

## 1.2 Installation

To use this package locally, ensure it's installed in your local Typst packages directory:

```
1 # Unix-like systems
2 ~/.local/share/typst/packages/local/gat-typst/0.1.0/
3
4 # Windows
5 %APPDATA%/typst/packages/local/gat-typst/0.1.0/
```

Then import it in your Typst documents:

```
1 #import "@local/gat-typst:0.1.0": *
```

## 1.3 Quick Start

Here's a minimal homework document to get started:

```
1 #import "@local/gat-typst:0.1.0": *
2
3 #show: homework.with(
4   title: "Homework 1",
5   course: "CS 101",
6   university: "Your University",
7   author: "Your Name",
8   email: "you@example.com",
9   date: datetime.today(),
10 )
11
12 #question(status: "done")[
13   Prove that  $P = NP$ .
14 ]
15
16 #solution[
```

```
17 Left as an exercise to the reader.
18
19 Using our custom math utilities:
20  $E(X^2) \geq (E(X))^2$  by Jensen's inequality.
21 ]
```

## 2 Important Behavior Changes

When using this package, several default behaviors are modified for improved mathematical typesetting and consistency.

### 2.1 Math Shorthands

The package enables extensive math operator shorthands via the `quick-maths` package. These are **automatically active** in all templates.

#### 2.1.1 Complete Shorthand List

Operators:

|   |                         |                          |       |
|---|-------------------------|--------------------------|-------|
| 1 | <code>\$a ** b\$</code> | // Small asterisk        | Typst |
| 2 | <code>\$a * b\$</code>  | // Centered dot $\Delta$ |       |
| 3 | <code>\$a +- b\$</code> | // Plus-minus            |       |
| 4 | <code>\$A  - B\$</code> | // Turnstile             |       |
| 5 | <code>\$a o+ b\$</code> | // Circled plus          |       |
| 6 | <code>\$a o. b\$</code> | // Circled dot           |       |
| 7 | <code>\$a @ b\$</code>  | // Also circled dot      |       |

$$a * b \quad a \cdot b \quad a \pm b$$

$$A \vdash B \quad a \oplus b$$

$$a \odot b \quad a \oslash b$$

Logic:

|   |                         |                   |       |
|---|-------------------------|-------------------|-------|
| 1 | <code>\$a .. b\$</code> | // Spacing (quad) | Typst |
| 2 | <code>\$A :: B\$</code> | // Therefore      |       |
| 3 | <code>\$A :' B\$</code> | // Because        |       |

$$a \quad b$$

$$A \therefore B \quad A \because B$$

Relations:

|   |                            |                         |       |
|---|----------------------------|-------------------------|-------|
| 1 | <code>\$a &lt;= b\$</code> | // Precedes or equals   | Typst |
| 2 | <code>\$a &lt; b\$</code>  | // Precedes             |       |
| 3 | <code>\$a &gt;= b\$</code> | // Succeeds or equals   |       |
| 4 | <code>\$a &gt; b\$</code>  | // Succeeds             |       |
| 5 | <code>\$a -= b\$</code>    | // Equivalent           |       |
| 6 | <code>\$a ~= b\$</code>    | // Asymptotically equal |       |
| 7 | <code>\$a \sim b\$</code>  | // Approximately        |       |

$$a \preceq b \quad a \prec b$$

$$a \succeq b \quad a \succ b$$

$$a \equiv b \quad a \cong b \quad a \approx b$$

Brackets:

|   |                                         |                       |       |
|---|-----------------------------------------|-----------------------|-------|
| 1 | <code>\$&lt;&lt; a, b &gt;&gt;\$</code> | // Angle brackets     | Typst |
| 2 | <code>\$a == b\$</code>                 | // Equals (unchanged) |       |

$$\langle a, b \rangle$$

$$a = b$$

#### 2.1.2 Warning About \* Shorthand

The `*  $\rightarrow$  dot.op` replacement is the most impactful change:

|   |                               |                                      |
|---|-------------------------------|--------------------------------------|
| 1 | // With shorthands (default): | Typst                                |
| 2 | <code>\$a * b\$</code>        | // Renders: $a \cdot b$              |
| 3 | <code>\$a ** b\$</code>       | // Renders: $a * b$ (small asterisk) |
| 4 |                               |                                      |
| 5 | // To use literal asterisk:   |                                      |

|                                                |
|------------------------------------------------|
| 6 <code>\$a ast b\$</code> // Renders: $a * b$ |
|------------------------------------------------|

$a \cdot b$  (renders as  $a \cdot b$ )

$a * b$  (small asterisk)

$a \cdot b$  (literal asterisk)

This follows conventions in linear algebra and physics but may surprise users expecting traditional asterisk multiplication.

### 2.1.3 Disabling Shorthands

If you need to disable shorthands:

```
1 #show: common.with(enable-shorthands: false)
```

Typst

Note: Currently this parameter exists but shorthands are applied regardless. This will be fixed in a future version.

### 2.1.4 Examples

```
1 // Statistics
2 $Ex(X) .. Var(X) = Ex(X^2) - (Ex(X))^2$
3
4 // Linear algebra
5 $A * B != B * A .. "in general"$
6
7 // Logic
8 $x > 0 :: x^2 > 0$
9
10 // Approximation
11 $e ~~ 2.718$
12
13 // Relations
14 $x <= y "iff" f(x) <= f(y)$
```

Typst

$$\mathbb{E}[X] \quad \text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$A \cdot B \neq B \cdot A \quad \text{in general}$$

$$x > 0 \therefore x^2 > 0$$

$$e \approx 2.718$$

$$x \preceq y \text{ iff } f(x) \leq f(y)$$

## 2.2 Superscript Transformations

Three special superscript transformations are enabled for common mathematical notation.

### 2.2.1 Transpose ( $\text{^T}$ )

```
1 $A^T = "transpose of" A$
2 $x^T y = \text{iprod}(x, y)$
3 $(A B)^T = B^T A^T$
```

Typst

$$A^T = \text{transpose of } A$$

$$x^T y = \langle x, y \rangle$$

$$(AB)^T = B^T A^T$$

Renders with proper transpose symbol ( $\text{^T}$ ) instead of superscript T.

### 2.2.2 Dagger ( $\text{^+}$ )

```
1 $A^+ = "Moore-Penrose
  pseudoinverse"$
2 $H^+ = "Hermitian transpose"$
```

Typst

$$A^{\dagger} = \text{Moore-Penrose pseudoinverse}$$

$$H^{\dagger} = \text{Hermitian transpose}$$

Renders as dagger symbol ( $\text{^{\dagger}}$ ).

### 2.2.3 Star (^\*)

```
1 $V^* = "optimal value function"$  Typst
2 $pi^* = "optimal policy"$
3 $x^* = \arg \min_x f(x)$
```

$V^*$  = optimal value function

$\pi^*$  = optimal policy

$x^* = \arg \min_x f(x)$

Renders with star symbol (★) instead of superscript asterisk.

This distinguishes from the asterisk operator and provides cleaner mathematical notation.

### 2.2.4 Interaction with Shorthands

The superscript transformations work correctly even when \* shorthand is enabled:

```
1 $A^T * B^*$ // Transpose A times  Typst
   optimal B
```

$A^T \cdot B^*$



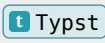
## 3 Utilities

The package provides extensive utilities for math, boxes, and display formatting.

### 3.1 Math Utilities

#### 3.1.1 Statistics Functions

Colored, bracketed notation for statistical operators:

|   |                               |                   |                                                                                   |
|---|-------------------------------|-------------------|-----------------------------------------------------------------------------------|
| 1 | <code>\$Ex(X)\$</code>        | // Expected value |  |
| 2 | <code>\$Var(X)\$</code>       | // Variance       |                                                                                   |
| 3 | <code>\$Cov(X, Y)\$</code>    | // Covariance     |                                                                                   |
| 4 | <code>\$Corr(X, Y)\$</code>   | // Correlation    |                                                                                   |
| 5 | <code>\$Pr(X &gt; 0)\$</code> | // Probability    |                                                                                   |

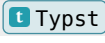
$\mathbb{E}[X]$     $\text{Var}[X]$     $\text{Cov}[X]$   
 $\text{Corr}[X]$     $\text{Pr}[X > 0]$

These render with blue coloring and square brackets. Example:

$\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$  by Jensen's inequality.

$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$  when  $X$  and  $Y$  are independent.

#### 3.1.2 Reasoning Operators

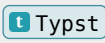
|   |                                                                                                                     |                |                                                                                   |
|---|---------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------|
| 1 | <code>\$x</code> <code>st</code> <code>x &gt; 0\$</code>                                                            | // "such that" |  |
| 2 | <code>\$If</code> <code>x &gt; 0</code> <code>Then</code> <code>y = 1</code> <code>Else</code> <code>y = 0\$</code> |                |                                                                                   |

$x$  s.t.  $x > 0$

if  $x > 0$  then  $y = 1$  else  $y = 0$

Example:  $\max\{x \text{ s.t. } x \in \mathbb{R}\}$

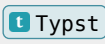
#### 3.1.3 Machine Learning / RL

|   |                           |                     |                                                                                     |
|---|---------------------------|---------------------|-------------------------------------------------------------------------------------|
| 1 | <code>\$KL(p, q)\$</code> | // KL divergence    |  |
| 2 | <code>\$refPol\$</code>   | // Reference policy |                                                                                     |

$D_{\text{KL}}(p \parallel q)$     $\bar{\pi}$

Example: Minimize  $D_{\text{KL}}(\pi \parallel \bar{\pi})$  to stay close to reference policy.

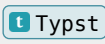
#### 3.1.4 Miscellaneous

|   |                           |                   |                                                                                     |
|---|---------------------------|-------------------|-------------------------------------------------------------------------------------|
| 1 | <code>\$SE\$</code>       | // Standard error |  |
| 2 | <code>\$ddot\$</code>     | // Double dot     |                                                                                     |
| 3 | <code>\$alias\$</code>    | // Alias arrow    |                                                                                     |
| 4 | <code>\$clamp(x)\$</code> | // Clamp function |                                                                                     |
| 5 | <code>\$solve(x)\$</code> | // Solve function |                                                                                     |

$\text{SE}$     $\ddot{\phantom{x}}$     $\overset{\text{alias}}{\leftarrow}$   
 $\text{clamp}\{x\}$     $\text{solve}\{x\}$

#### 3.1.5 Custom Math Function Factory

Create your own math functions with custom delimiters:

|   |                                                |                                                                                     |
|---|------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | <code>#let myFunc = math-func("myFunc",</code> |  |
|   | <code>delim: "[")</code>                       |                                                                                     |
| 2 | <code>\$myFunc(x, y)\$</code>                  |                                                                                     |
| 3 |                                                |                                                                                     |
| 4 | // Or with custom left/right delimiters        |                                                                                     |

```
#let norm = math-func("norm", ldelim:  
5 math.bar.v.double, rdelim:  
  math.bar.v.double)  
6 $norm(x)$
```

`myFunc[x]`

`norm||x||`

## 3.2 Box Utilities

### 3.2.1 gatbox()

Customizable colored boxes with automatic color cycling.

#### 3.2.1.1 Basic Usage

```
1 #gatbox[ t Typst
2   This is a basic box with default black
   color.
3 ]
4
5 #gatbox(color: blue)[
6   This is a blue box.
7 ]
```

This is a basic box with default black color.

This is a blue box.

#### 3.2.1.2 Color Cycling

Automatically cycle through predefined colors:

```
1 #gatbox(color-cycle: true)[First t Typst
   box - blue]
2 #gatbox(color-cycle: true)[Second box -
   red]
3 #gatbox(color-cycle: true)[Third box -
   green]
4 #gatbox(color-cycle: true)[Fourth box -
   orange]
5 #gatbox(color-cycle: true)[Fifth box -
   purple]
```

First box - blue

Second box - red

Third box - green

Fourth box - orange

Fifth box - purple

#### 3.2.1.3 Parameters

- `color` (color, default: `black`): Box color
- `color-cycle` (boolean, default: `false`): Auto-cycle through colors
- `title` (content): Optional title
- All `showybox` parameters supported

#### 3.2.1.4 Examples

```
1 #gatbox(title: "Important", t Typst
   color: red)[
2   This is a critical point!
3 ]
4
```

```
5 #gatbox(color-cycle: true, title: "Theorem
   1")[
6   All horses are the same color.
7 ]
```

Important

This is a critical point!

Theorem 1

All horses are the same color.

### 3.2.2 showybox

The base box system (re-exported from the `showybox` package) with `breakable: true` by default.

```
1 #showybox(t Typst  
2   title: "Note",  
3   frame: (  
4     border-color: blue,  
5     title-color: blue.lighten(80%),  
6   ),  
7 ) [  
8   Content here can break across pages.  
9 ]
```

Note

Content here can break across pages.

## 3.3 Display Utilities

### 3.3.1 Show Rules

The package includes custom show rules that can be used independently:

```
1 #show: super-ast-as-start Typst  
2  
3 $V^* = V(x^*)$ // Asterisk renders as  
   star symbol
```

$$V^* = V(x^*)$$

## 4 Common Styling

### 4.1 Code Styling

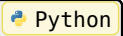
All code blocks are automatically styled:

#### 4.1.1 Automatic Changes

1. **Font size:** Reduced to 8pt for compactness
2. **Zebra striping:** Alternating row backgrounds (gray.lighten(80%))
3. **Border:** 1pt black stroke around code blocks
4. **Syntax highlighting:** Language-specific coloring

#### 4.1.2 Example

```
1 def quicksort(arr):  
2     if len(arr) <= 1:  
3         return arr  
4     pivot = arr[len(arr) // 2]  
5     left = [x for x in arr if x < pivot]  
6     middle = [x for x in arr if x == pivot]  
7     right = [x for x in arr if x > pivot]  
8     return quicksort(left) + middle + quicksort(right)
```



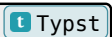
No manual configuration needed - just use standard Typst raw blocks with language tags.

### 4.2 Table Styling

The `booktabs` package is automatically applied, giving all tables professional styling with:

- Proper spacing between rows
- Professional horizontal rules (top, mid, bottom)
- No vertical lines (following `booktabs` philosophy)
- Better visual hierarchy

```
1 #table(  
2     columns: 3,  
3     [*Algorithm*], [*Time*], [*Space*],  
4     [Quicksort], [$O(n \log n)$], [$O(\log n)$],  
5     [Mergesort], [$O(n \log n)$], [$O(n)$],  
6     [Heapsort], [$O(n \log n)$], [$O(1)$],  
7 )
```



No special syntax required - standard `#table()` gets enhanced styling.

## 5 Templates

The package provides three main templates: `common`, `homework`, and `cheatsheet`. All templates are designed to work together and share common styling.

### 5.1 Common Template

The `common` template is the foundation that provides imports, styling, and configuration. It's automatically applied by both `homework` and `cheatsheet` templates, but can also be used standalone.

#### 5.1.1 Usage

```
1 #show: common.with()
```

 Typst

#### 5.1.2 Parameters

- `enable-shorthands` (boolean, default: `true`): Enable math operator shorthands

#### 5.1.3 Features Provided

The `common` template automatically configures:

1. **FontAwesome Icons:** Access to icon library via `fa-icon()` function
2. **Code Styling:** Syntax highlighting with Codly, 8pt font size, zebra striping
3. **Table Styling:** Professional booktabs-style tables by default
4. **Math Configuration:**
  - Equation numbering with `equate` package
  - Breakable equations
  - Sub-numbering disabled
5. **Show Rules:**
  - $\textsuperscript{T}$  renders as transpose
  - $\textsuperscript{+}$  renders as dagger ( $\dagger$ )
  - $\textsuperscript{*}$  renders as star ( $\star$ ) instead of asterisk

#### 5.1.4 Example

```
1 #import "@local/gat-typst:0.1.0": *
2
3 #show: common.with()
4
5 // Use FontAwesome icons
6 #fa-icon("heart") Math is beautiful
7
8 // Code with automatic styling
```

 Typst

```
python def hello(): print("Hello, world!")
```

```
1
2 // Professional tables
3 #table(
4   columns: 3,
5   [*Name*], [*Age*], [*City*],
6   [Alice], [25], [Boston],
```

```
7     [Bob], [30], [Seattle],  
8  )  
9  
10 // Math with special superscripts  
11 $ A^T dot B^+ = C^* $
```

## 5.2 Homework Template

The homework template creates professional homework assignments with custom components for questions, solutions, and more.

### 5.2.1 Usage

```
1  #show: homework.with(t Typst
2    title: "Homework 1",
3    course: "Course Name",
4    university: "University Name",
5    author: "Your Name",
6    email: "your@email.com",
7    collaborators: [List of collaborators],
8    resources: [Resources used],
9    date: datetime.today(),
10   cols: 1, // or 2 for two-column layout
11   show-header: true,
12   paper-size: "us-letter",
13 )
```

### 5.2.2 Parameters

- title (string): Homework title (e.g., “HW01”, “Problem Set 3”)
- course (content): Course name or code
- university (content): University name
- author (content): Your name
- email (content): Your email address
- collaborators (content, default: []): List of collaborators
- resources (content, default: []): Resources used
- date (datetime, default: `datetime.today()`): Assignment date
- cols (integer, default: 1): Number of columns (1 or 2)
- show-header (boolean, default: true): Show header with metadata
- paper-size (string, default: “us-letter”): Paper size

### 5.2.3 Components

#### 5.2.3.1 question()

Creates a colored box for problem statements with optional status tracking.

```
1  #question(status: "done")[t Typst
2    What is the derivative of  $x^2$ ?
3  ]
4
5  #question(status: "partial")[
6    This is partially complete.
7  ]
8
9  #question(status: "todo")[
```

```

10 Not started yet.
11 ]
12
13 #question()[
14 No status indicator.
15 ]

```

Status colors:

- "done": Green
- "partial": Yellow
- "todo": Red
- "none" or omitted: Black

### 5.2.3.2 solution()

Wrapper for solution content with optional column layout.

```

1 #solution[
2 The derivative is  $2x$ .
3 ]
4
5 #solution(cols: 2)[
6 This solution spans two columns.
7 ]

```

### 5.2.3.3 boxed-solution

Inline box for short solutions.

```

1 The answer is #boxed-solution[ $x = 5$ ].

```

### 5.2.3.4 sub-solution

Nested solution box for multi-part problems.

```

1 #sub-solution[
2 For part (a), we first note that...
3 ]

```

### 5.2.3.5 notes

Purple box for additional notes or remarks.

```

1 #notes[
2 Remember to check edge cases!
3 ]

```

## 5.2.4 Complete Example

```

1 #import "@local/gat-typst:0.1.0": *
2
3 #show: homework.with(

```



```

4   title: "HW02",
5   course: "Linear Algebra",
6   university: "MIT",
7   author: "Student Name",
8   email: "student@mit.edu",
9   collaborators: [Alice, Bob],
10  resources: [Textbook Chapter 3],
11  date: datetime(year: 2025, month: 1, day: 15),
12  cols: 1,
13 )
14
15 = Matrix Operations
16
17 #question(status: "done")[
18   Let  $A = \text{mat}(1, 2; 3, 4)$ . Compute  $A^T A$ .
19 ]
20
21 #solution[
22   We compute:
23   $
24    $A^T A = \text{mat}(1, 3; 2, 4) \text{mat}(1, 2; 3, 4)$ 
25    $= \text{mat}(10, 14; 14, 20)$ 
26   $
27
28   #sub-solution[
29     Note that  $A^T A$  is always symmetric.
30   ]
31 ]

```

## 5.3 Cheatsheet Template

The cheatsheet template creates compact, three-column cheatsheets perfect for exams.

### 5.3.1 Usage

```
1 #show: cheatsheet.with(t Typst
2   title: "Midterm",
3   course: "Course Name",
4   university: "University Name",
5   author: "Your Name",
6   email: "your@email.com",
7   date: datetime.today(),
8 )
```

### 5.3.2 Parameters

Similar to homework but without cols, show-header, and paper-size (fixed to 3-column layout on US Letter with 0.5cm margins).

### 5.3.3 Features

- **3-column layout** with 0.3cm gutter
- **Compact spacing**: 6pt base font, 7pt level-1 headings, 6pt level-2/3
- **No page numbers**: Maximizes space
- **Centered headings**: Better visual organization

### 5.3.4 Example

```
1 #import "@local/gat-typst:0.1.0": *t Typst
2
3 #show: cheatsheet.with(
4   title: "Final Exam Cheatsheet",
5   course: "Calculus I",
6   author: "Your Name",
7   date: datetime(year: 2025, month: 5, day: 10),
8 )
9
10 = Derivatives
11
12 == Power Rule
13 $ dv(, x) x^n = n x^(n-1) $
14
15 == Product Rule
16 $ dv(, x) (f g) = f' g + f g' $
17
18 == Chain Rule
19 $ dv(, x) f(g(x)) = f'(g(x)) g'(x) $
20
21 = Integrals
22
```

23 == **Power Rule**

24 \$ **integral** x^n **dd**(x) = x^(n+1)/(n+1) + C \$

## 6 Dependencies & Imports

The package integrates 20+ external packages, organized by category.

### 6.1 Diagramming

#### 6.1.1 CeTZ (v0.4.2)

Comprehensive drawing library for creating custom diagrams.

```
1 #import "@preview/cetz:0.4.2": canvas, draw
2
3 #canvas({
4   import draw: *
5   circle((0, 0), radius: 1)
6   line((0, 0), (1, 1))
7   content((0, 0), $x$)
8 })
```

Use for: Custom diagrams, network architectures, geometric figures.

#### 6.1.2 CeTZ Plot (v0.1.2)

Plotting extension for CeTZ.

```
1 #import "@preview/cetz-plot:0.1.2": plot
2
3 #canvas({
4   plot.plot(
5     size: (8, 6),
6     x-tick-step: 1,
7     y-tick-step: 1,
8     {
9       plot.add(x => x * x, domain: (-2, 2))
10    }
11  )
12 })
```

Use for: Function plots, data visualization.

#### 6.1.3 Fletcher (v0.5.8)

Arrow diagrams and state machines.

```
1 #diagram(
2   node((0, 0), $A$, radius: 2em),
3   node((1, 0), $B$, radius: 2em),
4   edge((0, 0), (1, 0), "->", $f$),
5 )
```

Use for: State machines, flowcharts, commutative diagrams.

### 6.1.4 Lilaq (v0.5.0)

Modern data visualization library.

```
1 #lq.diagram(t Typst  
2   title: [Data Plot],  
3   xlabel: $x$,  
4   ylabel: $y$,  
5   lq.plot((0, 1, 2, 3), (1, 4, 2, 3), mark: "o"),  
6 )
```

Use for: Scientific plots, cleaner syntax than CeTZ Plot.

## 6.2 Math

### 6.2.1 Matset (v0.1.0)

Set notation support.

Use for: Mathematical set operations and notation.

### 6.2.2 Equate (v0.3.2)

Advanced equation numbering with breakable equations.

```
1 $t Typst  
2   f(x) = x^2  
3 $ <eq:quadratic>  
4  
5 Reference: @eq:quadratic
```

$$f(x) = x^2$$

Use for: Equation numbering and cross-referencing. Automatically enabled.

### 6.2.3 Physica (v0.9.5)

Comprehensive physics notation package.

Key functions:

```
1 $vb(x)$           // Vectort Typst  
   bold  
2 $vu(u)$           // Unit vector  
3 $dd(x)$           // Differential  
4 $dv(f, x)$        // Derivative  
5 $pdv(f, x, y)$    // Partial  
6 $grad f$, $div vb(F)$, $curl vb(F)$  
7 $iprod(u, v)$      // Inner product  
8 $norm(x)$          // Norm  
9 $abs(x)$           // Absolute value  
10 $Set(x, x > 0)$   // Set builder  
11 $dmat(1, 2, 3)$    // Diagonal matrix  
12 $evaluated(f)_a^b$ // Evaluation bar
```

$$\begin{array}{l} \mathbf{x} \quad \hat{\mathbf{u}} \quad dx \\ \frac{df}{dx} \quad \frac{\partial^2 f}{\partial x \partial y} \\ \nabla f \quad \nabla \cdot \mathbf{F} \quad \nabla \times \mathbf{F} \\ \langle u, v \rangle \quad \|x\| \quad |x| \\ \{x \mid x > 0\} \\ \begin{pmatrix} 1 & & \\ & 2 & \\ & & 3 \end{pmatrix} \\ f|_a^b \end{array}$$

Example:

```

1 $
2 dv(vb(x), t) = vb(v), quad
3 pdv(f, x, y, [2, 1]) = 0, quad
4 iprod(vb(u), vb(v)) = norm(vb(u))
  norm(vb(v)) cos theta
5 $

```

$$\frac{dx}{dt} = v, \quad \frac{\partial^3 f}{\partial x^2 \partial y} = 0, \quad \langle u, v \rangle = \|u\| \|v\| \cos \theta$$

Use for: Almost all physics and advanced math notation.

#### 6.2.4 Quick-maths (v0.2.1)

Math shorthand system (powers the operator replacements).

Automatically enabled via common template. See section 5 for details.

#### 6.2.5 Mannot (v0.3.0)

Math annotations and labels.

TODO: FIX

```

1 $
2 p_i = exp(-beta E_i) / sum_j exp(-beta
  E_j)
3 #annot(<beta>, pos: top)[Inverse
  temperature]
4 $

```

Use for: Annotating complex equations, teaching materials.

### 6.3 Code

#### 6.3.1 Callisto (v0.2.4)

Jupyter notebook rendering.

```

1 #let (render, Cell, In, Out) = callisto.config(
2   nb: json("notebook.ipynb"),
3 )
4
5 #In("cell-tag") // Show input cell
6 #Out("cell-tag") // Show output cell
7 #Cell("cell-tag") // Show both

```

Use for: Including Jupyter notebook code/output in documents.

#### 6.3.2 Lovelace (v0.3.0)

Pseudocode formatting.

```

1 #pseudocode-list[
2   + *Input:* Array $A$ of length $n$
3   + *Output:* Sorted array
4   + *for* $i = 1$ *to* $n$ *do*
5     + Process element $A[i]$

```

```
6 ]
```

Use for: Algorithm descriptions.

### 6.3.3 Codly (v1.3.0) + Codly Languages (v0.1.1)

Modern syntax highlighting for code blocks.

Automatically configured with:

- 8pt font size
- 1pt black stroke
- Zebra striping (gray.lighten(80%))

```
1 def fibonacci(n):  
2     if n <= 1:  
3         return n  
4     return fibonacci(n-1) + fibonacci(n-2)
```

Python

Supports: Python, Rust, JavaScript, TypeScript, Java, C++, and 50+ more languages.

Use for: All code snippets. Automatically applied to raw blocks.

### 6.3.4 Cmarker (v0.1.6)

CommonMark (Markdown) support.

Use for: Converting Markdown to Typst.

## 6.4 Display & Formatting

### 6.4.1 Frame-it (v1.2.0)

Custom frame styles for content boxes.

```
1 #let (example, definition) = frames(  
2     example: ("Example", blue),  
3     definition: ("Definition", green),  
4 )  
5  
6 #example[Title][Tag][  
7     Body content here.  
8 ]
```

Typst

Use for: Theorem environments, examples, definitions.

### 6.4.2 FontAwesome (v0.6.0)

Font Awesome icon support (version 6).

```
1 #fa-icon("heart")  
2 #fa-icon("calculator")  
3 #fa-icon("robot")
```

Typst

Use for: Visual indicators, status icons. Note: Requires Font Awesome fonts installed.

### 6.4.3 Showybox (v2.0.4)

Styled, breakable boxes (base for gatbox).

```
1 #showybox(  
2   title: "Title",  
3   frame: (border-color: blue),  
4   breakable: true,  
5 ) [  
6   Content  
7 ]
```

Use for: Custom boxes, callouts, highlighted content.

### 6.4.4 Hydra (v0.6.2)

Context-aware page headers showing current section.

```
1 #set page(header: context [  
2   #text(fill: gray)[*#hydra(2)*]  
3 ])
```

Use for: Displaying current section in headers. Automatically enabled in homework template.

### 6.4.5 Booktabs (v0.0.4)

Publication-quality table styling.

Automatically applied to all tables. Creates professional spacing and rules.

Use for: All tables. No manual configuration needed.

## 6.5 Compatibility

### 6.5.1 MiTeX (v0.2.5)

LaTeX compatibility layer for including LaTeX code.

```
1 #mitex(`  
2   \begin{bmatrix}  
3     1 & 2 \\  
4     3 & 4  
5   \end{bmatrix}  
6 `)
```

Use for: Converting existing LaTeX, complex LaTeX-only constructs.

## 6.6 Color

### 6.6.1 Splash (v0.5.0)

Extended color support including xcolor names.

```
1 #text(fill: xcolor("red!50!blue"))[Mixed color]
```

Use for: Advanced color specifications.



### 6.6.2 Typpuccino (v0.1.0)

Catppuccin color schemes (frappe, latte, macchiato, mocha).

```
1 #import "@preview/typpuccino:0.1.0": mocha
2
3 #text(fill: mocha.red)[Catppuccin red]
```

 Typst

Use for: Consistent color theming.

## 7 Complete Examples

### 7.1 Simple Homework Document

```
1  #import "@local/gat-typst:0.1.0": *
2
3  #show: homework.with(
4    title: "Problem Set 1",
5    course: "Calculus II",
6    university: "Your University",
7    author: "Your Name",
8    email: "you@example.com",
9    date: datetime(year: 2025, month: 2, day: 1),
10 )
11
12 = Integration by Parts
13
14 #question(status: "done")[
15   Evaluate  $\int x e^x dx$ .
16 ]
17
18 #solution[
19   Using integration by parts with  $u = x$  and  $dv(x) = e^x dx$ :
20
21   
$$\int x e^x dx$$

22   
$$= x e^x - \int e^x dx \quad \backslash$$

23   
$$= x e^x - e^x + C \quad \backslash$$

24   
$$= e^x (x - 1) + C$$

25
26   
$$\int$$

27 ]
```

### 7.2 Math-Heavy Solution with Custom Functions

```
1  #import "@local/gat-typst:0.1.0": *
2
3  #show: homework.with(
4    title: "Statistics Assignment",
5    course: "Probability Theory",
6    author: "Student",
7    email: "student@school.edu",
8  )
9
10 = Random Variables
11
12 #question(status: "done")[
13   Let  $X$  and  $Y$  be independent random variables.
14   Show that  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ .
15 ]
```

```

15 ]
16
17 #solution[
18   By definition:
19   $
20   Var(X + Y) &= Ex((X + Y)^2) - (Ex(X + Y))^2 \
21   &= Ex(X^2 + 2X Y + Y^2) - (Ex(X) + Ex(Y))^2 \
22   &= Ex(X^2) + 2Ex(X Y) + Ex(Y^2) - (Ex(X))^2 - 2Ex(X)Ex(Y) - (Ex(Y))^2
23   $
24
25   Since $X$ and $Y$ are independent, $Ex(X Y) = Ex(X) * Ex(Y)$:
26   $
27   Var(X + Y) &= Ex(X^2) - (Ex(X))^2 + Ex(Y^2) - (Ex(Y))^2 \
28   &= Var(X) + Var(Y)
29   $
30
31 #sub-solution[
32   *Note:* This result generalizes: for independent $X_1, \dots, X_n$,
33   $
34   Var(sum_(i=1)^n X_i) = sum_(i=1)^n Var(X_i)
35   $
36 ]
37 ]

```

## 7.3 Compact Cheatsheet

```

1  #import "@local/gat-typst:0.1.0": *
2
3  #show: cheatsheet.with(
4    title: "Midterm Cheatsheet",
5    course: "Linear Algebra",
6    author: "Your Name",
7    date: datetime(year: 2025, month: 3, day: 15),
8  )
9
10 #let gatbox = gatbox.with(color-cycle: true)
11
12 = Matrix Operations
13
14 #gatbox(title: [Transpose])[
15   $(A^T)_(i j) = A_(j i)$
16
17   $(A B)^T = B^T A^T$
18 ]
19
20 #gatbox(title: [Determinant])[
21   $det(A B) = det(A) * det(B)$

```

```

22
23   $det(A^T) = det(A)$
24
25   $det(A^(-1)) = 1/det(A)$
26 ]
27
28 = Eigenvalues
29
30 #gatbox(title: [Definition])[
31   $A \, vb(v) = \lambda \, vb(v)$
32
33   $det(A - \lambda I) = 0$
34 ]
35
36 #gatbox(title: [Properties])[
37   $product_{i=1}^n \lambda_i = det(A)$
38
39   $sum_{i=1}^n \lambda_i = "tr"(A)$
40 ]
41
42 = Vector Spaces
43
44 #gatbox(title: [Subspaces])[
45   *Null space:* $cal(N)(A) = Set(vb(x), A \, vb(x) = vb(0))$
46
47   *Column space:* $cal(C)(A) = Set(A \, vb(x), vb(x) \text{ in } \mathbb{R}^n)$
48
49   *Row space:* $cal(C)(A^T)$
50 ]
51
52 #gatbox(title: [Fundamental Theorem])[
53   $dim \, cal(C)(A) + dim \, cal(N)(A) = n$
54
55   $cal(C)(A)^{perp} = cal(N)(A^T)$
56 ]

```

## 8 Tips & Best Practices

Based on conventions and optimal usage of this package:

### 8.1 Math Notation

1. Use **semantic symbols** instead of generic ones:

```
1 // Good
2  $\nabla f$  // Gradient
3  $\prod_{i=1}^n x_i$  // Product
4
5 // Avoid
6  $\nabla f$ 
7  $\prod_{i=1}^n x_i$ 
```

2. Space before subscripts with functions:

```
1 // Good
2  $V_\pi(s)$  // Value function of policy  $\pi$  at state  $s$ 
3  $Q_\theta(s, a)$ 
4
5 // Bad (no space before parenthesis)
6  $V_\pi(s)$ 
7  $Q_\theta(s, a)$ 
```

3. Use **physica utilities** for common notation:

```
1  $\mathbf{v}(x)$  // Vector bold
2  $\mathbf{u}(n)$  // Unit vector
3  $\mathcal{A}$  // Mathcal letters (script)
4  $\mathrm{d}x$  // Differential
5  $\mathrm{d}f(x)$  // Derivative
6  $\partial f(x, y)$  // Partial derivative
7  $\langle u, v \rangle$  // Inner product (not angle brackets)
8  $\|x\|$  // Norm (not  $||x||$ )
9  $|x|$  // Absolute value
10  $f|_a^b$  // Evaluation bar
```

4. Leverage **custom statistics functions**:

```
1 // These are colored and use square brackets
2  $\mathbb{E}(X)$  // Expectation
3  $\mathbb{V}(X)$  // Variance
4  $\mathbb{P}(A)$  // Probability
```

### 8.2 Layout

1. **Two-column homework** for dense problems:

```
1  $\#show: homework.with($ 
```

```

2   cols: 2,
3   // ... other params
4 )
5
6 // Override for specific solutions
7 #solution(cols: 1)[
8   This needs full width.
9 ]

```

2. **Color-cycle boxes** for organizing content:

```

1 #gatbox(color-cycle: true, title: "Theorem 1")[...]
2 #gatbox(color-cycle: true, title: "Theorem 2")[...]
3 #gatbox(color-cycle: true, title: "Theorem 3")[...]
4 // Automatically cycles: blue → red → green → orange → purple

```

3. **Status tracking** for homework progress:

```

1 #question(status: "done")[Completed problem]
2 #question(status: "partial")[Work in progress]
3 #question(status: "todo")[Not started]

```

## 8.3 Organization

1. Use **sub-solutions** for complex multi-part problems:

```

1 #solution[
2   Main solution approach...
3
4   #sub-solution[
5     For part (a), we have...
6   ]
7
8   #sub-solution[
9     For part (b), note that...
10  ]
11 ]

```

2. **Add notes** for clarifications or caveats:

```

1 #notes[
2   This assumes the matrix is invertible.
3   For singular matrices, use pseudoinverse.
4 ]

```

3. **Label equations** for cross-referencing:

```

1 $
2   f(x) = x^2 + 2x + 1
3 $ <eq:quadratic>

```

```
4
5 From @eq:quadratic, we see that...
```

## 8.4 Code

1. **Always specify language** for proper highlighting:

```
1 def example():
2     pass
```

 Python

Not just:

```
1 def example():
2     pass
```

2. **Use Callisto for notebook integration:**

```
1 #let (render, Cell, In, Out) = callisto.config(
2     nb: json("notebook.ipynb"),
3 )
4
5 // Tag cells in notebook, then reference them
6 #In("data-loading")
7 #Out("plot-results")
```

 Typst

## 8.5 Performance

1. **Compile time:** Heavy imports may slow compilation. Consider splitting large documents.
2. **Font Awesome:** Requires fonts installed system-wide. If icons don't render, install Font Awesome fonts.
3. **Cheatsheet margins:** 0.5cm is very tight. Adjust if printing issues occur:

```
1 #set page(margin: 0.7cm) // Before cheatsheet.with()
```

 Typst

## 8.6 Common Pitfalls

1. **Shorthand conflicts:** If you need literal `*` for multiplication:

```
1 $a ast b$ // Use 'ast' explicitly
```

 Typst

2. **Superscript confusion:** Remember `^*` becomes star:

```
1 $x^*$ // Renders as x* (star)
2 $x^ast$ // Renders as x* (asterisk if needed)
```

 Typst

3. **Column overrides:** Don't forget to override solution for full-width content in 2-column layouts:

```
1 #let solution = solution.with(cols: 1)
```

 Typst

## 9 Reference

### 9.1 Quick Import Guide

```
1 // Full import
2 #import "@local/gat-typst:0.1.0": *
3
4 // Selective import
5 #import "@local/gat-typst:0.1.0": homework, question, solution, gatbox
6
7 // Import specific utilities
8 #import "@local/gat-typst:0.1.0": Ex, Var, KL, vb, cal
```

### 9.2 Template Comparison

| Feature          | common  | homework                    | cheatsheet  |
|------------------|---------|-----------------------------|-------------|
| Columns          | N/A     | 1 or 2                      | 3 (fixed)   |
| Font size        | 11pt    | 13pt (1-col) / 10pt (2-col) | 6pt         |
| Headers          | None    | Optional                    | None        |
| Margins          | Default | 1 inch                      | 0.5cm       |
| Question boxes   | No      | Yes                         | No (manual) |
| Equation numbers | Yes     | Yes                         | Optional    |

### 9.3 Function Reference

#### 9.3.1 Templates

- `common(enable-shorthands: true, body)`
- `homework(title, course, university, author, email, ...)`
- `cheatsheet(title, course, university, author, email, ...)`

#### 9.3.2 Components

- `question(status: "none", ...)`
- `solution(cols: 1, body)`
- `boxed-solution`
- `sub-solution(...)`
- `notes(...)`

#### 9.3.3 Boxes

- `gatbox(color: black, color-cycle: false, ...)`
- `showybox(...)`

#### 9.3.4 Math (Statistics)

- `Ex()` - Expectation
- `Var()` - Variance
- `Cov()` - Covariance
- `Corr()` - Correlation



- `Pr()` - Probability

### 9.3.5 Math (Other)

- `KL(p, q)` - KL divergence
- `refPol` - Reference policy
- `st`, `If`, `Else`, `Then` - Logic
- `clamp(x)`, `solve(x)` - Functions

### 9.3.6 Utilities

- `date-format(date)` - Format datetime
- `math-func(name, ...)` - Create custom math function
- `super-ast-as-star` - Show rule for  $^*$

## 9.4 Further Reading

For detailed package documentation:

- **Physica**: <https://github.com/Leedehai/typst-physics>
- **CeTZ**: <https://cetz-package.github.io/docs/>
- **Fletcher**: <https://github.com/typst/packages/blob/main/packages/preview/fletcher/>
- **Codly**: <https://typst.app/universe/package/codly>
- **Typst Documentation**: <https://typst.app/docs/>

## 9.5 Version History

### v0.1.0 (Current)

- Initial release
- Templates: homework, cheatsheet, common
- 20+ integrated packages
- Math utilities and shorthands
- Custom box system

---

*For issues, suggestions, or contributions, see the project repository.*